

Polynomial removal fails for ordered binary matrices

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Abstract

We construct a fixed 66×66 binary matrix for which ordered matrix removal has no polynomial bound. This disproves the polynomial ordered binary matrix-removal conjecture. Ordered copies preserve the separate row and column orders and match both zeros and ones; removal permits changing entries in either direction.

1 Introduction

A binary matrix has both a row order and a column order. For a fixed binary $k \times k$ matrix H , an *ordered copy* of H in a binary $n \times n$ matrix A is a choice

$$r_1 < \cdots < r_k, \quad c_1 < \cdots < c_k$$

with $A(r_i, c_j) = H(i, j)$ for every $i, j \in [k]$, where $[k] = \{1, \dots, k\}$. Thus a copy matches the zeros as well as the ones, and the row and column choices are independent. Write $N_H(A)$ for the number of such copies and call A *H-free* if $N_H(A) = 0$. The normalized distance from H -freeness is

$$\text{dist}_H(A) = \frac{1}{n^2} \min_{B \text{ is } H\text{-free}} |\{(r, c) \in [n]^2 : A(r, c) \neq B(r, c)\}|,$$

where the minimum is over binary $n \times n$ matrices. Every cell may be changed, in either direction. We say that A is ϵ -far from being H -free when $\text{dist}_H(A) \geq \epsilon$.

The polynomial ordered binary matrix-removal conjecture asserts that for each fixed H there are constants $c_H, C_H > 0$ such that, for every integer $n \geq 1$, every $0 < \epsilon < 1$, and every binary $n \times n$ matrix A ,

$$N_H(A) \geq c_H \epsilon^{C_H} n^{2k} \quad \text{whenever} \quad \text{dist}_H(A) \geq \epsilon.$$

We disprove this conjecture with an explicit pattern of size 66.

Theorem 1.1. *There is a fixed binary 66×66 matrix H with the following property. For every integer $h \geq 1$, put*

$$n_h = (386h + 2)2^h, \quad \epsilon_h = (386h + 2)^{-2}.$$

There is a binary $n_h \times n_h$ matrix A_h such that

$$\text{dist}_H(A_h) \geq \epsilon_h, \quad \frac{N_H(A_h)}{n_h^{132}} \leq \epsilon_h 2^{-h}.$$

Consequently, no positive constants c, C give a polynomial removal bound for this fixed H .

The examples also give a lower bound for canonical sampling (Corollary 5.1). A tester that independently chooses uniform q -element row and column sets and rejects only when the sampled ordered submatrix contains H needs $q \geq \exp(\Omega(\epsilon_h^{-1/2}))$ for any fixed positive rejection probability along this sequence. Here q is the number of sampled positions on each axis; the submatrix has q^2 entries.

History and significance. Alon, Fischer and Newman raised the ordered-matrix question after proving polynomial-query testing for fixed finite forbidden families when row and column order is ignored [AFN07, Section 7]. Alon and Ben-Eliezer formulated the polynomial removal question for finite families of binary matrices [ABE20, Problem 1.4]. The singleton version appears explicitly as Conjecture 4.5 in the first arXiv version of the survey by Gishboliner and Shapira [GS25]. Theorem 1.1 resolves that conjecture negatively and answers the finite-family question negatively already for one square pattern.

Qualitative ordered matrix removal was established by Alon, Ben-Eliezer and Fischer [ABEF17, Theorem 1.8 in the full version]: a positive copy-density bound exists for each fixed positive distance. Our examples concern its quantitative dependence. Polynomial bounds also hold for finite families closed under row permutations, or under column permutations by transposition [ABE20, Theorem 1.5]. The polynomial induced removal classification for ordered graphs of Gishboliner and Tomon [GT22, Theorem 1.1] concerns a different model, with a single vertex order and symmetric adjacency.

Nonpolynomial lower bounds were already known for ternary hosts: Fischer and Rozenberg obtained them for a fixed two-symbol 2×2 pattern, with row and column order ignored [FR07]. Alon and Ben-Eliezer give a related ternary-matrix construction based on progression-free sets [ABE20, Section 5.1]. These examples use a third symbol in the host matrix; our construction uses only 0 and 1.

A particularly relevant positive result is the polynomial lower bound for matrices containing many entry-disjoint copies of a fixed pattern [ABE20, Theorem 1.2]. It yields a small set of cells meeting all existing copies when the copy density is small. Changing those cells can create new copies. Our construction makes this distinction explicit: all original copies meet a set of 2^h cells, while any binary repair changes at least 4^h cells.

Proof strategy. The last two rows and columns of H meet in the pattern

$$P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Its first 64 rows and columns specify which pairs of rows and columns can play these two roles. The host matrix contains many copies of this 64×64 anchor, each imposing a different instance of the same 2×2 restriction. Section 2 gives the pattern and the host entries explicitly.

The host has rows and columns indexed by a binary tree, with a plus block and a minus block at each internal node on each axis. At each leaf, the two blocks on each axis coincide. One root entry is 1 and the other is 0. Avoiding P in the designated bodies propagates these values down a common path, first to the next row level and then to the next column level. At the leaves the two signs share their positions, so they cannot retain opposite values.

We first prove the distance bound in Section 3. Fix an arbitrary edited matrix and sample a tree path together with anchor positions and auxiliary “dummy” positions, whose entries against the tree positions are all 1 in the original host. Each axis is partitioned into classes of equal size $m = 2^h$. For every cell needed to keep the sampled anchors, dummy incidences and root values unchanged, its row and column are selected with probability m^{-2} . Summing over the actual edited cells shows that fewer than m^2 changes leave a suitable selection. All other entries may have changed; avoidance of H itself forces the contradictory propagation in the edited matrix. This cellwise sampling argument avoids a loss depending on the number of classes and gives the inverse-square distance in Theorem 1.1.

The copy bound concerns the unedited host and requires locating every copy, beyond the designated anchors used in the distance proof. In Section 4, dense rows and columns, together with

a five-column block whose rows realize all binary words, force every anchor copy to occupy one of the prescribed anchor blocks. The remaining rows and columns must then have that block's designated roles. The order and threshold rules then force every copy to use a diagonal leaf cell, giving the small copy count. Section 5 combines the two estimates and derives the sampling consequence.

2 A fixed pattern and its tree matrices

We specify a binary pattern of size 66 and an ordered binary matrix A_h for each integer $h \geq 1$. The pattern consists of a 64×64 block and two additional rows and columns. The matrices A_h place copies of that block before positions indexed by a binary tree.

2.1 The pattern

Set $s = 64$. For $0 \leq a < 32$ and $1 \leq b \leq 5$, let

$$\eta_b(a) = \left\lfloor \frac{a}{2^{5-b}} \right\rfloor \bmod 2,$$

so $(\eta_1(a), \dots, \eta_5(a))$ lists the five-bit words in lexicographic order. Define $S \in \{0, 1\}^{[s] \times [s]}$ by

$$S(u, v) = \begin{cases} \eta_{v-59}(u-33), & 33 \leq u \leq 64, \quad 60 \leq v \leq 64, \\ \mathbf{1}[u \neq v], & \text{otherwise.} \end{cases}$$

Let e_1, e_2 be the first two standard basis vectors of $\mathbb{R}^s$. The fixed pattern and its lower-right block are

$$H = \begin{pmatrix} S & e_1 & e_2 \\ e_1^\top & 1 & 0 \\ e_2^\top & 1 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad k = s + 2 = 66. \quad (2.1)$$

We call S the *anchor* and P the *body* of H .

2.2 The ordered positions

Fix $h \geq 1$ and put $m = 2^h$. At depth i the full binary tree has nodes indexed by $p \in \{0, \dots, 2^i - 1\}$, with children $2p, 2p + 1$. For $0 \leq i < h$, each node has two disjoint row blocks $R_i^-(p), R_i^+(p)$ and two disjoint column blocks $C_i^-(p), C_i^+(p)$, each of size $m/2^i$. At depth h it has one row block and one column block of size 1; we give each of these two names:

$$R_h^-(p) = R_h^+(p), \quad C_h^-(p) = C_h^+(p).$$

Apart from these named equalities, all row blocks are disjoint, and all column blocks are disjoint. Their positions are called *variable* positions. Write

$$R_i^\sigma = \bigcup_{p=0}^{2^i-1} R_i^\sigma(p), \quad C_i^\sigma = \bigcup_{p=0}^{2^i-1} C_i^\sigma(p), \quad \sigma \in \{-, +\}.$$

Each of these sets has size m . For $1 \leq i \leq h$ and $d \in \{0, 1\}$, the sets $R_{i,d}^\sigma, C_{i,d}^\sigma$ are the corresponding unions restricted to node indices $p \equiv d \pmod{2}$. Introduce also a dummy row block R_* and a dummy column block C_* , each of size m .

Order the variable rows recursively. At an internal node, place its minus block, then its two subtrees in child order 0, 1, and then its plus block. At a leaf place its single block once. Order

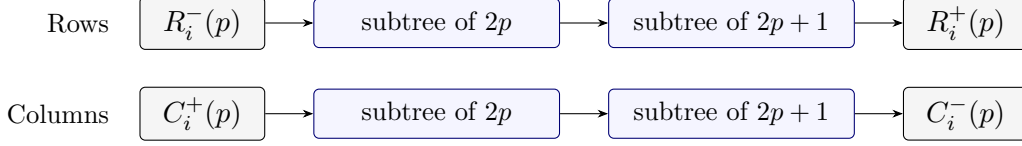


Figure 1: The two orders at an internal node (i, p) , where $i < h$. Each blue box contains the entire corresponding subtree on that axis. At a leaf, the minus and plus names denote the same single position. The reversed parent signs give the four comparisons in (2.2).

variable columns by the same recursion with the signs reversed: the plus block comes before the subtrees and the minus block after them. Place R_* after all variable rows and C_* before all variable columns. Within each block choose any fixed order.

For two disjoint blocks, $X < Y$ means that every position of X precedes every position of Y on their common axis. Let $\pi(p) = \lfloor p/2 \rfloor$ be the parent index of a nonroot node. For $1 \leq i \leq h$, indices p, q at depth i , and indices a, b at depth $i-1$, the recursive orders give

$$\begin{aligned} R_i^+(p) < R_{i-1}^+(a) &\iff \pi(p) \leq a, & R_{i-1}^-(a) < R_i^-(p) &\iff \pi(p) \geq a, \\ C_{i-1}^+(b) < C_i^+(q) &\iff \pi(q) \geq b, & C_i^-(q) < C_{i-1}^-(b) &\iff \pi(q) \leq b. \end{aligned} \quad (2.2)$$

Indeed, the subtrees rooted at depth $i-1$ occur consecutively in increasing node order. A child block lies between the two blocks of its own parent, including when that child is a leaf. Comparing its parent index with a or b proves all four equivalences. Figure 1 displays the two recursive orders.

Two elementary restrictions explain the body roles we use. For binary values a, b ,

$$\begin{pmatrix} 1 & a \\ 1 & b \end{pmatrix} \neq P \iff b \leq a, \quad \begin{pmatrix} a & b \\ 1 & 1 \end{pmatrix} \neq P \iff a \leq b.$$

We will supply the fixed ones with a dummy column in the first array and a dummy row in the second. The first restriction passes a value between consecutive row levels; the second passes it between consecutive column levels. In either inequality, knowing that the smaller entry is 1 forces the larger entry to be 1, while knowing that the larger entry is 0 forces the smaller entry to be 0.

To specify which rows and columns have these roles, define a set T of $6h$ indices, called *modes*. Each mode specifies two row sets and two column sets as follows, where $1 \leq i \leq h$ and $d \in \{0, 1\}$:

t	$R_1(t)$	$R_2(t)$	$C_1(t)$	$C_2(t)$
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
$W_{i,d}^+$	$R_{i,d}^+$	R_*	C_{i-1}^+	$C_{i,d}^+$
$W_{i,d}^-$	$R_{i,d}^-$	R_*	$C_{i,d}^-$	C_{i-1}^-

(2.3)

The V modes pair consecutive row levels with a dummy column; the W modes pair consecutive column levels with a dummy row and fix the child parity. For each t , its two row sets are disjoint and its two column sets are disjoint. An ordered body in mode t uses rows $r_1 < r_2$ with $r_j \in R_j(t)$ and columns $c_1 < c_2$ with $c_j \in C_j(t)$. The order conditions remain part of this definition; the two role sets need not precede one another as entire sets.

Fix the order of modes by increasing i , with the six modes at depth i ordered as

$$V_i^+, \quad V_i^-, \quad W_{i,0}^+, \quad W_{i,1}^+, \quad W_{i,0}^-, \quad W_{i,1}^-.$$

Before all variable and dummy positions on each axis, place an anchor block for each mode, in this mode order. The row block of mode t has successive groups $\widehat{R}_t(u)$, $1 \leq u \leq s$, and its column block has successive groups $\widehat{C}_t(v)$, $1 \leq v \leq s$. Each group has size m . Positions in these groups are *anchor positions*; the variable and dummy positions are *non-anchor positions*.

There are $6hs$ anchor groups on each axis. The variable positions form $2h + 1$ disjoint sets of size m : two signs at each depth below h , and one set of leaves. Including the dummy block, the total number of positions on either axis is therefore

$$n = d_h m, \quad d_h = 6hs + 2h + 2 = 386h + 2. \quad (2.4)$$

Identify each resulting ordered axis with $[n]$.

2.3 All binary entries

Define $A_h \in \{0, 1\}^{[n] \times [n]}$ by the following rules. They assign every entry, with no extra labels attached to the matrix.

1. Between anchor groups $\widehat{R}_t(u)$ and $\widehat{C}_{t'}(v)$, put $S(u, v)$ if $t = t'$ and put 0 otherwise.
2. For a non-anchor row r , its values against the s anchor column groups of mode t are $e_j^\top$ if $r \in R_j(t)$, and the zero word if it belongs to neither row set. Similarly, for a non-anchor column c , its values against the anchor row groups of mode t are e_j if $c \in C_j(t)$, and zero if it belongs to neither column set. The disjointness of the two role sets makes both instructions unambiguous.
3. Between non-anchor positions, put 1 whenever the row or column is dummy. Between variable positions, use the following table, putting 0 in every case absent from the table:

row block	column block	depth range	A_h
$R_i^+(p)$	$C_i^+(q)$	$0 \leq i \leq h$	$\mathbf{1}[p \leq q]$
$R_i^-(p)$	$C_i^-(q)$	$0 \leq i < h$	$\mathbf{1}[p < q]$
$R_i^+(p)$	$C_{i-1}^+(q)$	$1 \leq i \leq h$	$\mathbf{1}[\pi(p) \leq q]$
$R_i^-(p)$	$C_{i-1}^-(q)$	$1 \leq i \leq h$	$\mathbf{1}[\pi(p) < q]$

(2.5)

Every displayed value is constant on the corresponding pair of blocks. The first two lines of (2.5) have disjoint ranges of blocks because the second excludes the shared leaves. In the last two lines the column depth is below h , so their two column signs remain distinct even when the row is a shared leaf. Thus the table has no conflicting assignments. In particular, the entry between leaf blocks of indices p, q is always $\mathbf{1}[p \leq q]$.

3 Distance under arbitrary binary changes

We first prove that fewer than m^2 binary changes cannot eliminate all copies of H . We choose one path through the tree on each axis, with the same node indices, so that its root entries and sampled anchor and dummy incidences are unchanged. All other entries between variable positions may change freely. If the edited matrix were H -free, its mode restrictions would then force opposite values on the shared leaf entry.

Proposition 3.1. *Let $h \geq 1$, let $m = 2^h$, and let $A = A_h$ be the matrix constructed above, of order $n = d_h m$, where $d_h = 386h + 2$. Every H -free binary $n \times n$ matrix differs from A in at least m^2 entries. Consequently A is d_h^{-2} -far from being H -free.*

Proof. Fix an arbitrary binary matrix B differing from A on a set E of $M < m^2$ cells. We will find a copy of H in B .

Choosing positions while avoiding the relevant changes. On each axis, partition the positions into the anchor groups, the dummy group, the variable classes at levels $0, \dots, h-1$ for each sign, and the one shared leaf class. All these classes have size m . In particular, the leaf class occurs once in this partition, despite its two sign names.

Choose z uniformly from $\{0, \dots, m-1\}$ and put

$$a_i = \left\lfloor \frac{z}{2^{h-i}} \right\rfloor \quad (0 \leq i \leq h).$$

Thus a_i indexes the ancestor at level i of leaf z , and $\lfloor a_i/2 \rfloor = a_{i-1}$. Choose positions

$$x_i^\sigma \in R_i^\sigma(a_i), \quad y_i^\sigma \in C_i^\sigma(a_i) \quad (\sigma \in \{+, -\}),$$

uniformly within their blocks, using a single shared choice on each axis at the leaf: $x_h^+ = x_h^-$ and $y_h^+ = y_h^-$. Also choose one uniform position from every anchor group and from each dummy group; denote the dummy choices by x_* and y_* . Conditional on z , all choices in distinct groups are independent, including choices on opposite axes.

Each variable choice is uniform on its size- m class. Indeed, a fixed node at depth i is selected with probability 2^{-i} , and its block has size $m/2^i$, so each position in that class is selected with probability $1/m$. An anchor or dummy choice is independent of every choice on the opposite axis. The root choices are also independent uniform choices in their fixed size- m blocks. It follows that, for every cell (r, c) having at least one anchor or dummy position, or having both positions at the root level,

$$\Pr(r \text{ and } c \text{ are selected}) = \frac{1}{m^2}.$$

There is only one choice in each size- m class, so each cell occurs in this calculation once. Summing over the changed cells of these types gives

$$\Pr(\text{some selected cell of these types belongs to } E) \leq \frac{|E|}{m^2} < 1. \quad (3.1)$$

Fix a selection avoiding all such changes. Thus all selected anchor and dummy incidences, and the selected root entries, agree in A and B . Changes at every other cell remain unrestricted.

What a mode forbids. For each mode t , its selected anchor positions form the matrix S in B . They precede all selected non-anchor positions, and their entries to the selected positions in $R_j(t)$ and $C_j(t)$ are the prescribed vectors e_j . Therefore any ordered pair of selected rows in $R_1(t), R_2(t)$ and ordered pair of selected columns in $C_1(t), C_2(t)$ whose entries in B form

$$P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

complete these anchors to a copy of H . Suppose, for a contradiction, that B is H -free. Then this body matrix cannot occur for any mode.

Propagating the root values. The unchanged root entries give

$$B(x_0^+, y_0^+) = 1, \quad B(x_0^-, y_0^-) = 0. \quad (3.2)$$

We claim that the same two equalities hold at every level:

$$B(x_i^+, y_i^+) = 1, \quad B(x_i^-, y_i^-) = 0 \quad (0 \leq i \leq h). \quad (3.3)$$

Assume they hold at level $i-1$, where $1 \leq i \leq h$. For V_i^+ use rows x_i^+, x_{i-1}^+ and columns y_*, y_{i-1}^+ . For V_i^- use rows x_{i-1}^-, x_i^- and columns y_*, y_{i-1}^- . These are in their designated sets and in the required order because the chosen nodes are parent and child. Their body matrices in B are respectively

$$\begin{pmatrix} 1 & B(x_i^+, y_{i-1}^+) \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & B(x_i^-, y_{i-1}^-) \end{pmatrix}.$$

The dummy incidences supply the left column of ones. Avoiding P forces

$$B(x_i^+, y_{i-1}^+) = 1, \quad B(x_i^-, y_{i-1}^-) = 0.$$

To carry these values to level- i columns, put $d = a_i \bmod 2$. For $W_{i,d}^+$ use rows x_i^+, x_* and columns y_{i-1}^+, y_i^+ ; for $W_{i,d}^-$ use rows x_i^-, x_* and columns y_i^-, y_{i-1}^- . Again these positions have the required order and membership. The resulting matrices are

$$\begin{pmatrix} 1 & B(x_i^+, y_i^+) \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} B(x_i^-, y_i^-) & 0 \\ 1 & 1 \end{pmatrix}.$$

Here the bottom rows are unchanged dummy incidences. Avoiding P forces exactly (3.3) at level i , completing the induction.

At level h , the two asserted values in (3.3) concern the same cell, since the leaf positions were shared. This contradiction proves that B contains H . Since every matrix with fewer than m^2 changes has this property and $n = d_h m$, the normalized distance is at least $m^2/n^2 = d_h^{-2}$. $\square$

4 Copies are confined to the leaf diagonal

The distance proof used the prescribed anchor representatives. To bound the number of copies in the unedited matrix A_h , we must now locate every possible copy. Its 64×64 prefix will lie in the anchor groups of a single mode, and its two remaining rows and columns will then belong to that mode's designated sets.

4.1 The anchor and the non-anchor traces

The anchor matrix S has two complementary features. Its lower-right corner contains every binary word of length five. We will show that non-anchor rows cannot realize all these words on five non-anchor columns. This will force more than 32 positions on at least one axis of any copy to be anchors. The dense first 32 rows and columns, contrasted with the sparse variable-to-anchor incidences and the zero entries between different modes, will then force the whole copy into one mode.

Lemma 4.1 (Properties of the anchor). *The rows of S are pairwise distinct, as are its columns. Every column has at most one zero in the first 32 rows, and every row has at most one zero in the first 32 columns. Each row has at least 58 ones and each column has at least 48 ones. Finally, the restrictions of rows 33, ..., 64 to columns 60, ..., 64 are the 32 distinct binary words of length five.*

Proof. The replacement in the definition of S does not meet the first 32 rows or the first 59 columns. In the first 59 columns, each of rows $1, \dots, 59$ has its own distinguishing zero, while rows $60, \dots, 64$ consist entirely of ones. The latter five rows are distinct in the replacement block. Thus the rows are distinct, and each has at least 58 ones.

Each of the first 59 columns has exactly one zero, in its own row. The last five columns are distinct because they are distinct coordinates on the list of all binary words of length five. Each has exactly 16 zeros, all in the replacement block, so it also differs from every earlier column and has 48 ones. The remaining assertions follow directly from the location and definition of that block. $\square$

Lemma 4.2 (Incidence with the modes). *A variable position on either axis belongs to designated sets of at most four modes. Consequently, against any collection containing at most one representative of each anchor group on the opposite axis, a variable position has at most four entries equal to one.*

Proof. Consider a non-leaf row in $R_i^\sigma(p)$. The only possible modes are V_i^σ , V_{i+1}^σ , and $W_{i,d}^\sigma$, where $d = p \bmod 2$; modes whose level is outside $1, \dots, h$ are omitted. A leaf row belongs to V_h^+ , V_h^- , and the two modes $W_{h,d}^+$, $W_{h,d}^-$. There are at most four in either case.

For a non-leaf column in $C_i^\sigma(p)$, the possible modes are V_{i+1}^σ , $W_{i+1,0}^\sigma$, $W_{i+1,1}^\sigma$, and $W_{i,d}^\sigma$, again omitting undefined levels. A leaf column belongs only to $W_{h,d}^+$ and $W_{h,d}^-$. For each participating mode, the entries to its anchor groups form one unit vector. This proves the last assertion as well. $\square$

A set of columns is *shattered* by a set of rows if the restrictions of those rows to the columns include every binary word of the corresponding length.

Lemma 4.3 (No five non-anchor columns are shattered). *For any five distinct non-anchor columns of A_h , the non-anchor rows do not realize all 32 binary words on those columns.*

Proof. If one of the five columns is a dummy column, its entry is one in every non-anchor row, so the conclusion is immediate. Otherwise all five columns are variable. Partition them according to their classes C_i^σ , counting the common leaf class only once.

Within each such class, order the selected columns by node index (and arbitrarily within a node block). The entry rules for A_h show that the ones of a variable row form a suffix in each class in which that row is nonzero. A non-leaf row is nonzero in at most two classes: its own level and sign, and the preceding level with the same sign. A leaf row is nonzero in at most three classes: the common leaf class and the two signs at level $h - 1$.

A class containing a selected columns has at most a distinct nonempty suffixes. Across all classes there are therefore at most five such suffixes. The trace of a variable row is a union of at most three of them, giving at most

$$\sum_{j=0}^3 \binom{5}{j} = 26$$

possible traces. The dummy rows contribute at most one additional trace. Thus at most $27 < 32$ words occur. $\square$

4.2 Every anchor copy uses one mode

Lemma 4.4 (Anchor rigidity). *Every ordered copy of S in A_h uses the row and column anchor groups of a single mode t . On each axis it uses exactly one position from each of that mode's 64 groups, in their prescribed order.*

Proof. Fix an ordered copy of S . Two positions in a single anchor group have identical entries throughout A_h . By Lemma 4.1, the copy therefore uses at most one position from any one anchor group. The same reasoning shows that it uses at most one dummy position on each axis.

Suppose that at most 32 selected rows and at most 32 selected columns are anchors. Since all anchors precede all non-anchor positions, the last 32 selected rows and the last five selected columns are non-anchor positions. Their entries give all 32 binary words by Lemma 4.1, contrary to Lemma 4.3. Hence more than 32 selected positions are anchors on at least one axis.

Assume first that more than 32 selected rows are anchors. In particular the first 32 selected rows are anchors. No selected column can be variable: Lemma 4.2 would give at most four ones in these rows, whereas each column of S has at least 31 ones in its first 32 rows. There are consequently at least 63 selected anchor columns, since at most one selected column is dummy.

Any two selected anchor columns have a common one among the first 32 selected rows: each has at most one zero there. They must therefore belong to the same mode, because entries between anchor groups of different modes are zero. Write t for this common mode. Every selected anchor row also belongs to t . Indeed an anchor row of another mode would be zero on the at least 63 selected anchor columns and would have at most one one in the entire copy, contradicting Lemma 4.1.

A selected dummy column is now impossible: its entries to the selected anchor rows of mode t contain at most one one, because its entries to that mode form either a unit vector or the zero vector. This again contradicts the density of the first 32 rows of S . Thus all selected columns are anchors of mode t . Any non-anchor row has at most one one against those columns, so every selected row must also be an anchor of t .

If instead more than 32 selected columns are anchors, the same argument applies with the axes interchanged. All properties used after the preceding case distinction are symmetric in the axes: entries between different anchor modes are zero, the first 32 rows and columns of S each contain at most one zero in the relevant restriction, all its rows and columns have more than one one, and Lemma 4.2 holds on both axes.

There are 64 selected positions on each axis and only 64 anchor groups in mode t . No group is repeated, so every group is used. Their order in the construction is their order in the copy. $\square$

Corollary 4.5 (The two body roles). *In every ordered copy of H in A_h , the first 64 rows and columns are anchor representatives of one mode t . Its two body rows belong respectively to $R_1(t)$, $R_2(t)$, and its two body columns belong respectively to $C_1(t)$, $C_2(t)$.*

Proof. Apply Lemma 4.4 to the copy's prefix S . The two body rows must have entries $e_1^\top$ and $e_2^\top$ against these anchor columns. An anchor row of another mode gives the zero vector; an anchor row of mode t gives a row of S , which has more than one one. Thus neither body row is an anchor. By the definition of the entries between non-anchor positions and anchor groups, the vector $e_j^\top$ occurs exactly on $R_j(t)$, for $j = 1, 2$. The two designated sets are disjoint. The column argument is identical. $\square$

4.3 Every copy uses a diagonal leaf cell

The body restrictions from Corollary 4.5 leave only one possible source of copies: the two signs share the same positions at the leaves. We now locate that source at a fixed entry of the pattern.

Proposition 4.6. *For every integer $h \geq 1$, every ordered copy of H in A_h maps the pattern entry $(s+1, s+1)$ into*

$$\mathcal{L}_h = \bigcup_{p=0}^{m-1} R_h^+(p) \times C_h^+(p).$$

In particular, $|\mathcal{L}_h| = m$ and

$$N_H(A_h) \leq mn^{2k-2}, \quad \frac{N_H(A_h)}{n^{2k}} \leq \frac{2^{-h}}{d_h^2}. \quad (4.1)$$

Proof. By Corollary 4.5, the last two rows and columns of an H -copy lie in the prescribed body sets of a single mode. Their entries must form

$$P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

We check the four kinds of modes, using the order equivalences (2.2) and the entry rules (2.5). Write $\pi(p) = \lfloor p/2 \rfloor$ for the parent index of a node.

In mode V_i^+ , let p be the child-level row index, a the parent-level row index, and b the parent-level column index. The order of the two rows gives $\pi(p) \leq a$. The bottom-right entry of P gives $a \leq b$. Consequently $\pi(p) \leq b$, so the top-right entry is 1, a contradiction.

In mode V_i^- , use the same names for the node indices at the same levels. The row order gives $\pi(p) \geq a$, while the top-right entry 0 gives $a \geq b$. Thus $\pi(p) \geq b$, and the bottom-right entry is 0, again a contradiction.

In mode $W_{i,d}^+$, let p, q be the child-level row and column indices, and let b be the parent-level column index. The column order gives $b \leq \pi(q)$, and the top-left entry 1 gives $\pi(p) \leq b$. Since $p \equiv q \equiv d \pmod{2}$, it follows that $p \leq q$. The top-right entry is therefore 1, contrary to P .

Finally, in mode $W_{i,d}^-$, the column order gives $\pi(q) \leq b$, and the top-right entry 0 gives $\pi(p) \geq b$. The common parity implies $p \geq q$. If $i < h$, the top-left entry is $\mathbf{1}[p < q] = 0$, which is impossible. If $i = h$, that entry is instead $\mathbf{1}[p \leq q]$, because the leaf blocks are shared. It can be 1 only when $p = q$. This places the top-left entry of the body, namely $(s+1, s+1)$ in H , in $\mathcal{L}_h$.

Each leaf block has size $m/2^h = 1$, so $\mathcal{L}_h$ contains exactly m cells. For each such cell, fix the images of row $s+1$ and column $s+1$ of H . There are at most n^{k-1} choices for the other row indices and at most n^{k-1} choices for the other column indices; requiring increasing orders and the prescribed entries can only reduce these numbers. This proves the first inequality in (4.1). Since $n = d_h m$ and $m = 2^h$, division by n^{2k} gives the second. $\square$

Remark 4.7. Changing the m entries of $\mathcal{L}_h$ from 1 to 0 destroys every original copy, but creates new ones. Indeed, fix a leaf index p and put $d = p \bmod 2$. In mode $W_{h,d}^+$, choose the leaf row in $R_h^+(p)$ and any dummy row. Choose, in this order, a column in $C_{h-1}^+(\pi(p))$ and the leaf column in $C_h^+(p)$. These rows and columns have the required order. Their body is now

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} :$$

the upper-left entry is the unchanged parent-child entry, the upper-right entry is the changed leaf diagonal, and the lower entries are dummy incidences. Choosing one position from every anchor group of this mode completes a new copy of H , since all anchor entries and signatures are unchanged. Thus an original-copy hitting set need not provide an H -free repair.

5 Proof of the main theorem

The construction uses the same 66×66 pattern H at every depth. The two preceding estimates now give the required counterexamples.

Proof of Theorem 1.1. For each integer $h \geq 1$, set

$$\epsilon_h = d_h^{-2} = (386h + 2)^{-2}, \quad n_h = d_h 2^h.$$

Proposition 3.1 shows that every H -free binary matrix differs from A_h in at least $m^2 = \epsilon_h n_h^2$ entries. Proposition 4.6 gives

$$\frac{N_H(A_h)}{n_h^{132}} \leq \epsilon_h 2^{-h}.$$

These statements prove the asserted construction sequence.

Now fix arbitrary real numbers $c, C > 0$. As h tends to infinity,

$$\frac{N_H(A_h)}{\epsilon_h^C n_h^{132}} \leq (386h + 2)^{2C-2} 2^{-h} \rightarrow 0.$$

Choose an integer h for which the right-hand side is less than c . Then $0 < \epsilon_h < 1$, the matrix A_h is ϵ_h -far from being H -free, and $N_H(A_h) < c \epsilon_h^C n_h^{132}$, as required. $\square$

Corollary 5.1 (Canonical row-column sampling). *Fix $h \geq 1$ and an integer $0 \leq q \leq n_h$. Consider a tester that chooses a uniformly random q -element row set $R \subseteq [n_h]$ and, independently, a uniformly random q -element column set $C \subseteq [n_h]$. It rejects only if the induced submatrix $A_h[R, C]$, with the inherited orders, contains an ordered copy of H . Then*

$$\Pr(\text{reject on } A_h) \leq \epsilon_h 2^{-h} q^2,$$

and the rejection probability is zero when $q < 66$. For every fixed $\rho \in (0, 1]$, rejection probability at least ρ requires

$$q \geq \sqrt{\rho} \epsilon_h^{-1/2} 2^{h/2},$$

hence $q \geq \exp(\Omega(\epsilon_h^{-1/2}))$ along the sequence ϵ_h as $h \rightarrow \infty$. Here q is the sample size on each axis; the induced submatrix has q^2 entries.

Proof. By Proposition 4.6, every copy of H in A_h uses a cell of $\mathcal{L}_h$, and $|\mathcal{L}_h| = 2^h$. For each fixed cell (r, c) , independence gives $\Pr((r, c) \in R \times C) = (q/n_h)^2$. Rejection implies that $R \times C$ meets $\mathcal{L}_h$, so the union bound gives

$$\Pr(\text{reject on } A_h) \leq 2^h (q/n_h)^2 = \epsilon_h 2^{-h} q^2.$$

If $q < 66$, the sampled submatrix cannot contain H . Rearranging the probability bound gives the necessary sample size, and $\epsilon_h^{-1/2} = 386h + 2$ gives its exponential growth along the stated sequence. $\square$

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